Inverse z-transform

Practice questions · Day 15 · Sat 7 Nov · pack D8

5. Practice

From your course files

T7 Q1 (tutorial). Determine all possible signals $x(n)$ with the following z-transform:

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

T7 Q2 (tutorial). Determine the signal $x(n)$ with the z-transform:

$$X(z) = \frac{3}{1-\frac{10}{3}z^{-1}+z^{-2}}$$

if $x(n)$ is absolutely summable.

T7 Q3 (tutorial). A *causal* discrete-time LTI system produces the output

$$y(n) = \left(\frac{1}{3}\right)^n u(n)$$

in response to the input signal

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1).$$

(a) Determine the impulse response $h(n)$ and transfer function $H(z)$ of this system.
(b) Find the difference equation that characterises the system.
(c) Is this system stable?

T8 Q5(c) (tutorial, optional redo). A communication channel is modelled by the transfer function

$$H(z) = \frac{1 + 0.7z^{-1} - 1.2z^{-2}}{1 - 0.8z^{-1} + 0.64z^{-2}}, \qquad \text{ROC: } |z| > 0.8$$

(c) Determine the transfer function and impulse response of its stable inverse. Is it causal, anticausal or noncausal?

New questions

Q1. Power series [4 marks]

a. $X(z) = 2z^2 - z + 3 - 4z^{-2}$, ROC $0 < |z| < \infty$. Write $x(n)$ as a sequence with the $n = 0$ sample marked. [1]

b. $X(z) = \dfrac{1 + 2z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$. Use long division to find $x(0), \ldots, x(3)$ if $x(n)$ is causal. [2]

c. Same $X(z)$, but now the ROC is $|z| < 1/\sqrt{2}$. Find $x(-1)$ and $x(-2)$. [1]

Q2. LCCDE, improper $H(z)$ and system types [7 marks]

Three causal systems:

- S1: $y(n) - 0.5y(n-1) = x(n) + x(n-1)$
- S2: $y(n) = x(n) - \frac{1}{8}x(n-3)$
- S3: $y(n) = 0.9y(n-2) + x(n)$

a. Find $H_1(z)$ and its poles and zeros. [2]

b. Find $h_1(n)$ in closed form. [3]

c. Classify each system as FIR (all-zero), all-pole or pole-zero. Give its poles, and say whether it is stable. [2]

Q3. Find all possible sequences [7 marks]

$$X(z) = \frac{1 + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})}$$

a. Expand $X(z)$ in partial fractions. [3]

b. List every possible $x(n)$ with its ROC, and label each causal, anti-causal or two-sided. [3]

c. Which one has a DTFT? Check your answer by evaluating $X(1)$ two ways. [1]

Q4. System ID → LCCDE → stability [10 marks]

A causal LTI system has input $x(n) = u(n)$ and output

$$y(n) = \left[\frac{2}{3} - \frac{1}{5}\left(\frac{1}{2}\right)^n + \frac{8}{15}(-2)^n\right]u(n)$$

a. Find $X(z)$ and $Y(z)$ as single fractions, with ROCs. [2]

b. Find $H(z)$ and the difference equation. [3]

c. Find $h(n)$. Is the system stable? [2]

d. Another system obeys the same difference equation but is known to be stable. Find its $h(n)$. Is it causal? [2]

e. Would the system in (d) produce the same $y(n)$ for the input $u(n)$? Give a one-line reason. [1]

Q5. Improper $H(z)$ with complex poles [8 marks]

$$H(z) = \frac{1 + z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2}}, \qquad \text{causal}$$

a. Find the poles and zeros. Is the system stable? [2]

b. Use long division to find $h(0), \ldots, h(3)$. [2]

c. Find $h(n)$ in real closed form. Check it against (b). [3]

d. What is the steady-state output for $x(n) = \cos(\pi n/2)$? [1]

6. Quick recall check

1. Invert $\dfrac{4}{1 - 0.8z^{-1}}$ with ROC $|z| < 0.8$.
2. A causal system has poles at 0.9 and -1.1. Is it stable?
3. $X(z)$ has poles at 0.5, 2 and 5. How many possible $x(n)$ are there, and which ROC gives the absolutely summable one?
4. Write the LCCDE for $H(z) = \dfrac{1 + 2z^{-1}}{1 - 0.25z^{-2}}$.
5. $H(z) = 1 - 2z^{-1} + z^{-3}$. What type of system is it, where are its poles, and is it stable?